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-- cursors
-- A cursor in MySQL is a database object that allows you to retrieve and process
rows
-- on at a time from a result set.
-- like a pointer that moves through query results row by row.
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-- syntax:-
-- 1st step :- declare the cursor:-
declare cursor_name cursor for select ....;
-- 2nd step :- declare the handler for when no more rows exists
declare continue handler for not found set done = true;
-- 3rd step:- open the cursor
open cursor_name;
-- 4th step - fetch rows in loops
while not done do
    fetch cursor_name into variable1, variable1, ...;
    -- process here
end while;
-- 5th step :- close the cursor
close cursor_name;
```

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create database cursors;
use cursors;
```

```
CREATE TABLE employees (
    emp_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100),
    salary DECIMAL(10,2),
    bonus DECIMAL(10,2)
);
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```
INSERT INTO employees (name, salary, bonus) VALUES
('Amit Sharma', 50000, 5000),
('Neha Verma', 60000, 6000),
('Rahul Mehta', 45000, 4000),
('Sneha Kapoor', 70000, 8000),
('Vikas Singh', 55000, 4500),
('Pooja Gupta', 48000, 3500),
('Rohan Das', 62000, 7000),
('Anjali Nair', 53000, 3000),
('Karan Patel', 75000, 9000),
('Meera Iyer', 68000, 6500),
('Arjun Reddy', 72000, 7200),
('Priya Malhotra', 51000, 2800),
('Suresh Yadav', 47000, 3200),
('Divya Bansal', 66000, 7500),
('Manish Jain', 59000, 5000),
('Kavita Joshi', 61000, 6200),
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('Deepak Chauhan', 54000, 4100),
('Shreya Sen', 73000, 8500),
('Nitin Agarwal', 56000, 4600),
('Isha Khanna', 64000, 6800);

update employees set bonus = null;

select * from employees;
-- give different bonus % based on salary:-
-- salary < 30000 then 20% bonus
-- salary 30000- 60000 then 10% bonus
-- salary > 60000 then 5% bonus

delimiter $$
create procedure calculate_bonus()
begin
    declare done boolean default false;
    declare v_id int; -- 1
    declare v_salary decimal(9,2);
    declare v_bonus decimal(9,2); -- 5000

    -- cursor
    declare emp_cursor cursor for
    select emp_id, salary from employees; -- 1 50000
    -- handler
    declare continue handler for not found
    set done = true;

    open emp_cursor;

    read_loop : loop
    fetch emp_cursor into v_id, v_salary; -- 2 60000

    if done then
    leave read_loop;
    end if; -- exit

    -- bussiness logic
    if v_salary < 30000 then
    set v_bonus = v_salary * 0.2;
    elseif v_salary <= 60000 then
    set v_bonus = v_salary * 0.1; -- 6000
    else
    set v_bonus = v_salary * 0.05;
    end if;

    update employees
    set bonus = v_bonus -- 6000
    where emp_id = v_id; -- 2

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        end loop read_loop;

        close emp_cursor;
end $$
delimiter ;
```

```
select * from employees;

call calculate_bonus();
```